Summary of activities, 2 March 2026

Plan & deliverables: At the outset of this week, I met with Bin to establish a plan for the week. We determined that I would focus on three primary deliverables for this week, detailed below: (1) primer designs targeting *GBA1* N370S; (2) a graphical abstract for the Familial PD INTERACT document; (3) a battery of tests to evaluate *GBA1* correction *in vitro*. Additional focuses included establishing long-term goals of increasing my fluency with gene editing and iPSC reprogramming techniques and literature, as well as completing my onboarding.

1. N370S primer designs

I designed two primers targeting the *GBA1* N370S mutation based on project parameters that we established: an amplicon length between 200–700 bp incorporating intronic differences between *GBA1* and the pseudogene *GBAP1* to facilitate their bioinformatic distinction downstream. I designed these primers using NCBI primer-BLAST; the exact method and final primers are detailed in the attached document. With Bin's approval, I emailed these to Kiara (David Liu Lab) for her feedback and ordered these, along with Kiara's L444P primers, from GeneWiz.

2. INTERACT graphical abstract

I consulted with Bin to draft a suitable graphical abstract for incorporation into the Familial PD INTERACT document. We determined that hewing to Bin's original design on the project workflow made sense, with a particular point of emphasis on the one main addition to this workflow (for which we are seeking approval): the gene editing component ("One-step strategy"). Once Bin had approved my draft, I rendered the final image by hand and assembled the figure in PowerPoint to ensure it will be easily manipulable for future presentations. My final edit ensured that the depiction of the reprogramming factors is more identifiably mRNA per my discussion with Bin. The final figure is attached at the end of this document.

3. *In vitro* battery for *GBA1* correction

I designed a battery of tests to verify recovery of functional deficits following *GBA1* correction *in vitro* based on a survery of the literature. These tests, along with other candidate tests, are summarized in the attached document.

4. Cell culture

I also began my training to culture fibroblasts and hiPSCs with Bin and Tatyana. I observed media changes last week and on Monday, and on Tuesday I observed Tatyana passaging two 6-cm. dishes of PD05 mR01 hiPSCs. On Wednesday, I passaged PD05 mR03 hiPSCs 1:1 into two 6-cm. dishes under Bin and Tatyana's observation. I independently conducted media changes for these two lines on Thursday and Friday (as detailed on my LabArchives, with the accompanying worksheets). I also acted as verifier for Bin while she passaged and cryopreserved fibroblasts on Tuesday and prepared samples for karyotyping (00W2D2 DAPs; PD05-eB01R11-P20 and eB01R42-P2 hiPSCs) on Friday.

5. Other tasks

I completed my onboarding process, including troubleshooting my Outlook mailbox, email address domain, and card access to the lab. I corrected the hiPSC passaging worksheet (as a separate copy on Teams) and placed several orders (Staples: binders, flash drives; LabSuit: P/S, MEM-NEAA).

1. Objective and criteria: These primers are to validate the presence and genetic correction of the *GBA1* (ID: 2629) N370S (nucleotide 1226) mutation in patient fibroblasts. They should amplify a <700 bp amplicon of the gene, ideally one that is bioinformatically distinguishable from the *GBAP1* pseudogene. Since we know the L444P primers Kiara designed should meet these criteria, they can serve as a reference.

The location of the N370S mutation is c.1226 at position 155235843 (on GRCh38).

2. Method: To generate these primers, I used NCBI primer-BLAST to search NC_0000011.1 between positions 155235400 and 155236200 to generate 200–700 bp amplicons with a minimum melting temperature of 59°C. The search summary:

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Input PCR template: NC_000001.11 Homo sapiens chromosome 1, GRCh38.p14
Primary assembly range: 155235400-155236200
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I used nucleotide-BLAST to compare the FASTA sequences for this region against the *GBAP1* pseudogene sequence (ID: AF267177.1), which accords with the validated sequence (ID: 2630).

Note that, as with Kiara's primers, I have indicated point mutations in red with the pathogenic mutation underlined and highlighted the exons in gray.

Option 1:

Forward (5′–3′): GGTTCCACCCTGAACACCTT
Reverse (5′–3′): CACACCCCCAACTCCTTAGC

GBA1:
CACACCCCCAACTCCTTAGCTACTAAGGAATGTGCCCCTCACAGGGCTGA
CCTACCCACAGCTGCCTCTCCCACATGTGACCCTTACCTACACTCTCTGG
GGACCCCCAGTGTTGCGCCTTTGTCTCTTTGCCTTTGTCCTTACCCTAGA
GCCTCCTGTACCATGTGGTCGGCTGGACCGACTGGAACCTTGCCCTGAAC
CCCGAAGGAGGACCCAATTGGGTGCGTAACTTTGTCGACAGTCCCATCAT
TGTAGACATCACCAAGGACACGTTTTACAAACAGCCCATGTTCTACCACC
TTGGCCACTTCAGGTGAGTGGAGGGCGGGCACCCCCATTCCATACCAGGC
CTATCATCTCCTACATCGGATGGCTTACATCACTCTACACCACGAGGGAG
CAGGAAGGTGTTCAGGGTGGAACC

GBAP1:
CACACCCCCAACTCCTTAGCTACTAAGGAATGTGCCCCTCACAGGGCTGA
CCTACCCACAGCTGCCTCTCCCACACGTGACCCTTACCTACACTCTCTGG
GGACCCCCAGTGTTGAGCCTTTGTCTCCTTGCCTTTGTCCTTACCCTAGA
ACCTCCTGTACCATGTGGTCGGCTGGACCGACTGGAACCTTGCCCTGAAC
CCCGAAGGAGGACCCAATTGGGTGCGTAACTTTGTCGACAGTCCCATCAT
TGTAGACATCACCAAGGACACGTTTTACAAACAGCCCATGTTCTACCACC
TTGGCCACTTCAGGTGAGTGGAGGGCGGGCACCCCCATTCCATACCAGGC
CTATCATCTCCTACATCGGATGGCTTACATCACTCTACACCACGAGGGAG
CAGGAAGGTGTTCAGGGTGGAACC

Sequence (5′–3′)	Length	Start	Stop	T_m	GC%	Self complementarity	Self 3′ complementarity
GGTTCCACCCTGAACACCTT	20	155235570	155235589	59.81	55	4	0
CACACCCCCAACTCCTTAGC	20	155235993	155235974	60.32	60	3	2

Length: 424 bp

Option 2:

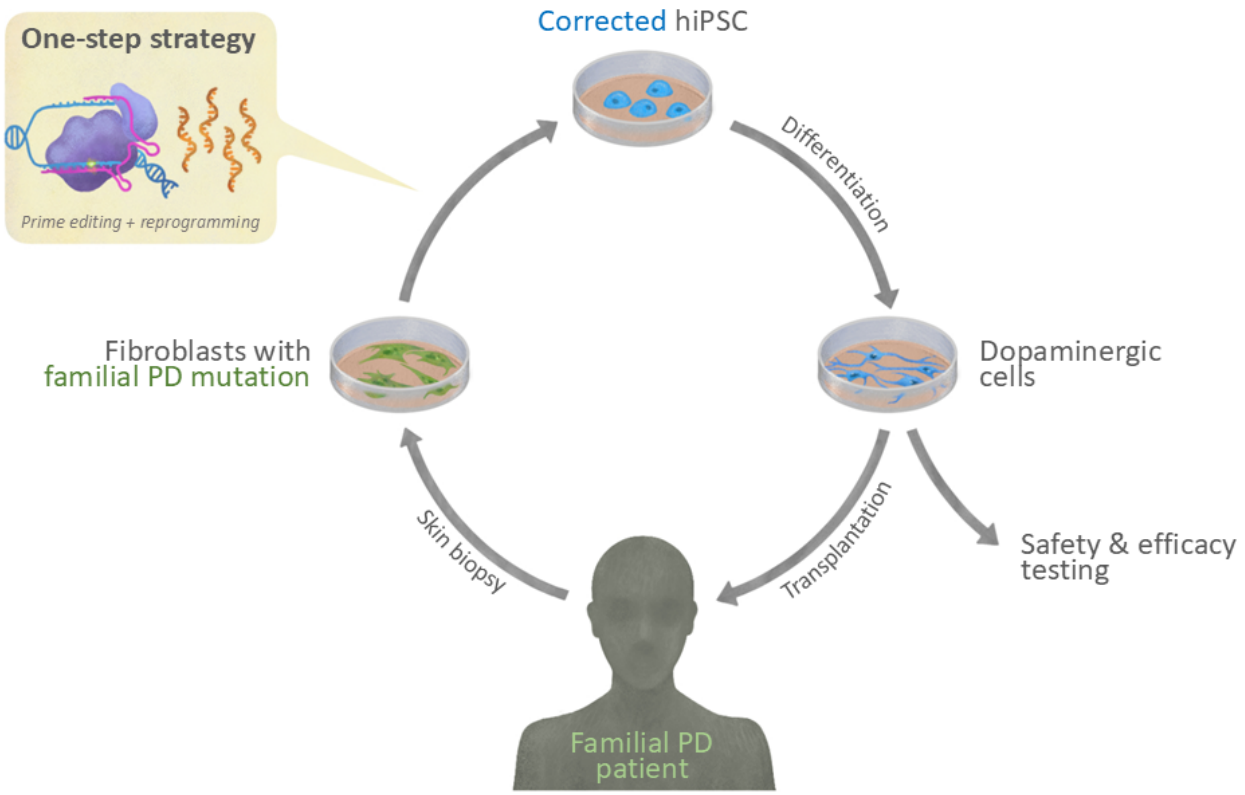
Forward (5′–3′): GTCGACAAAGTTACGCACCCAAT
Reverse (5′–3′): CTACTAAGGAATGTGCCCCTCA

GBA1:
CTACTAAGGAATGTGCCCCTCACAGGGCTGACCTACCCACAGCTGCCTCT
CCCACATGTGACCCTTACCTACACTCTCTGGGGACCCCCAGTGTTGCGCC
TTTGTCTCTTTGCCTTTGTCCTTACCCTAGAGCCTCCTGTACCATGTGGT
CGGCTGGACCGACTGGAACCTTGCCCTGAACCCCGAAGGAGGACCCAATT
GGGTGCGTAACTTTGTCGAC

GBAP1:
CTACTAAGGAATGTGCCCCTCACAGGGCTGACCTACCCACAGCTGCCTCT
CCCACACGTGACCCTTACCTACACTCTCTGGGGACCCCCAGTGTTGAGCC
TTTGTCTCCTTGCCTTTGTCCTTACCCTAGAACCTCCTGTACCATGTGGT
CGGCTGGACCGACTGGAACCTTGCCCTGAACCCCGAAGGAGGACCCAATT
GGGTGCGTAACTTTGTCGAC

Sequence (5′–3′)	Length	Start	Stop	T_m	GC%	Self complementarity	Self 3′ complementarity
GTCGACAAAGTTACGCACCCAAT	23	155235755	155235777	61.9	47.83	6	2
CTACTAAGGAATGTGCCCCTCA	22	155235974	155235953	59.23	50	3	1

Length: 202 bp


One-step strategy
Prime editing + reprogramming
Corrected hiPSC
Differentiation
Dopaminergic cells
Safety & efficacy testing
Transplantation
Familial PD patient
Skin biopsy
Fibroblasts with familial PD mutation

1. Project objective and criteria: Construct a battery of tests to demonstrate recovery of functional defects in patient-derived hiPSC dopaminergic neurons following correction of *GBA1* mutation(s) *in vitro*. Stem cell–derived neurons harboring *GBA1* L444P and N370S mutations have been extensively characterized (see Schondorf et al., 2014; Aflaki et al., 2016; Grigor'eva et al., 2023). Here, I have given preference to robust, reliably demonstrated phenotypes, especially those demonstrated in L444P or N370S neurons or across multiple cellular models. I have included an estimated Cohen's d based on reported means and standard deviations.

The final battery of tests is briefly outlined below; the following section (2.) contains a rationale, method, and citations for each test.

Measure	**Method**	**Expectation***	**Est. effect size**	**Citations**
Lysosomal-autophagic flux	Western blot: LC3II, actin; + NH_4Cl, leupeptin	↑	$d \approx 6$	Mazzulli et al., 2011; Mubariz et al., 2023; Schondorf et al., 2014
GCase enzyme activity	MUB-Glc + CBE; total protein	↑	$d \approx 4$	Schondorf et al., 2014; Mubariz et al., 2023; Grigor'eva et al., 2023
Lysosomal–GCase co-localization	ICC: LAMP1 and GCase	↑	$d \approx 6$	Ivanova et al., 2011; Navarro-Romero et al., 2022
Lipid droplet accumulation	ICC: BODIPY	↓	$d \approx 1.75$	Smith et al., 2023; Nuber et al., 2025
α-synuclein accumulation	Western blot: αS, phospho-αS, actin or GAPDH	↓	$d \approx 4$	Schondorf et al., 2014; Mazzulli et al., 2011; Navarro-Romero et al., 2022
and aggregation	+ PFFs	↓	$d \approx 2.5$	Kim et al., 2018; Gegg et al., 2020
Rate of apoptosis	Live plate reader	↓	$d \approx 3.25$	Perez-Abshana et al., 2024

* *Relative to mutant.*

Other tests include: lysosomal number and size, GCase maturation, lysosomal accumulation of lipid substrates, and ER stress. In many cases, these are redundant with my proposed battery (e.g., GCase maturation and lysosomal–GCase co-localization), but all would be reasonable to include.

2. Phenotypes, methods, and rationale:

2.1. Lysosomal-autophagic flux

Rationale: Human *GBA1*-Parkinson's and Gaucher's disease dopaminergic neurons display decreased proteolysis [1-3]. Notably, loss of GCase function also interrupts the function of other lysosomal enzymes, providing more evidence for its general disruption of lysosomal-autophagy function [1].

Method: Western blot: LC3II, actin; + NH_4Cl and leupeptin [2, 3]; radioactive pulse-chase [1]

Sources cited: [1] Mazzulli et al., 2011 ($d \approx 8$); [2] Mubariz et al., 2023 ($d \approx 5$); [3] Schondorf et al., 2014 ($d \approx 4.75$).

2.2. GCase enzyme activity

Rationale: GCase enzme activity is perhaps the most obvious phenotype to look for as a consequence of mutations in its encoding gene, and so it's also one of the best-supported phenotypes. Importantly, decreased GCase activity has been observed in *GBA1*-mutant Gaucher's and Parkinson's disease patient-derived iPSC midbrain dopaminergic neurons [1, 4] and dopaminergic-like neuroblastoma carrying either the *GBA1* L444P and N370S mutations [2], among a variety of other animal models. Loss of GCase function also reduces the effectiveness of other lysosomal enzymes [3–4].

Method: MUB-Glc + CBE (GCase inhibitor), normalized to total protein (several kits are available for purchase)

Sources cited: [1] Schondorf et al., 2014 ($d \approx 5$); [2] Mubariz et al., 2023 ($d \approx 2.5$) [3] Grigor'eva et al., 2023 ($d \approx 4.25$); [4] Mazzulli et al., 2011.

2.3. Lysosomal–GCase co-localization

Rationale: Correct trafficking of GCase to the lysosome is necessary for its function and its disruption with *GBA1* (and other lysosomal proteins, i.e. *PSAP*) have been repeatedly demonstrated across various *GBA1*-Parkinson's disease models. Notably, GCase is glycosylated as it is transported through the Golgi, meaning that its "maturation" is detectable ratiometrically by Western blot; however, a rigorous demonstration of this defect requires EndoH/PNGase digestion for validation. Here, I have given preference to co-localization as a much simpler read-out.

Method: ICC: GCase, LAMP1

Sources cited: [2] Ivanova et al., 2021; [3] Navarro-Romero et al., 2022 ($d \approx 6$).

2.4. Lipid droplet accumulation

Rationale: Lipid droplets are a general indicator of loss of lipid homeostasis, which has been variably reported with *GBA1* mutations associated with Parkinson's disease. These include the accumulation of GCase substrates [Mazzulli et al., 2011] as well as monounsaturated lipid species [2], among others. Lipid droplet accumulation has also been linked to Parkinson's disease more generally, both in patient brain samples (see Brekk et al., *PNAS*, 2020) and various model systems (see Girard et al., *PLoS Genetics* 2021). Increased lipid droplet number has been demonstrated in *GBA1* L444P, N370S, and E326K neuroblastoma [2] and lipid droplet number is modestly increased in *GBA1* L444P and E326K mouse brain [1].

Method: BODIPY (neutral lipid probe), Plin2 (lipid droplet marker)

Sources cited: [1] Nuber et al., 2025 ($d \approx 1.5$); [2] Smith et al., 2023 ($d \approx 2$).

2.5. α-synuclein accumulation

Rationale: αS is increased in *GBA1* N370S and L444P-harbouring patient-derived iPSC midbrain dopaminergic neurons [1]. In my own work, I have also

observed total and phosphorylated αS to be elevated in *GBA1* L444P and E326K patient-derived iPSC cortical neurons [2].

Method: Western blot: αS (4B12 or Syn1), p-αS, and GAPDH

Sources cited: [1] Schondorf et al., 2014 ($d \approx 3.5$); [2] Mazzulli et al., 2011 ($d \approx 2.5$); Navarro-Romero et al., 2022 ($d \approx 6.25$).

2.6. α-synuclein aggregation

Rationale: αS aggregation with PFFs has been demonstrated in *GBA1* N370S hiPSC-derived dopaminergic neurons [1], E326K primary glial cells (Kweon et al., 2024), and organoids (Frattini et al., 2025). Notably, Mazzulli et al. (2011) suggests that αS toxicity occurs through a misfolding- (and potentially aggregation-) dependent process, as primary neurons expressing the aggregation-incompetent Δ71–82 αS variant do not display αS-associated toxicity. Please note that this study did not use PFFs.

Method: PFF treatment

Sources cited: Kim et al., 2018 ($d \approx 3.25$); Gegg et al., 2020 ($d \approx 2$).

2.7. Rate of apoptosis

Rationale: Decreased cell viability is another consistently reported phenotype in *GBA1*-mutant cellular models, including dopaminergic-like neuroblastoma carrying either *GBA1* L444P or N370S mutations (Navarro-Romero et al., 2022). *GBA1* K198E skin fibroblasts exhibit greater cleaved caspase 3 [1]. Altered cell viability and resilience has been demonstrated by a number of different methods.

Method: Live plate reader (I am anticipating using the same method as Kritika has planned for the *PRKN* cells)

Sources cited: [1] Perez-Abshana et al., 2024.

2.8. Lysosomal number and size

Rationale: Altered lysosomal morphology has been consistently observed with *GBA1* mutations in Parkinson's disease models [1–3]. Given that lysosomal function is a more functional readout, I have favored that measure; however, lysosomal size, number, morphology, pH, and biogenesis is easily observed.

Method: ICC: LAMP1; LysoTracker

Sources cited: [1] Navarro-Romero et al., 2022; [2] Peng et al., 2021; [3] Nuber et al., 2025; [4] Schondorf et al., 2014.

2.9. ER stress

Rationale: Misfolded GCase is hypothesized to be retained in the ER rather than trafficked through the Golgi and to the lysosome; this, in addition to the accumulation of lipid substrates, causes ER stress and subsequently activation of the unfolded protein response (UPR). Decreased GCase maturation leading to increased UPR activation and decreased lysosomal–GCase co-localization has been demonstrated in L444P and N370S dopaminergic-like neuroblastoma [1].

Method: Western blot: IRE1α, ATF6, p-elF2, CHOP, actin

Sources cited: Navarro-Romero et al., 2022.

2.10. GCase maturation

Rationale: See 2.3. Lysosomal–GCase co-localization and 2.9. ER stress for my rationale in excluding this read-out. I have previously demonstrated decreased GCase maturation in *GBA1* L444P and E326K iPSC-derived cortical neurons.

Method: Western blot: GCase, actin; +EndoH, +PNGase

Sources cited: [1] Mazzulli et al., 2011; [2] Navarro-Romero et al., 2022; [3] Nuber et al., 2025.

2.11. Lysosomal accumulation of lipid substrates

Rationale: Accumulation of lipid substrates is consistently observed with *GBA1* mutations in Gaucher's disease, but reports in Parkinson's disease brain are conflicting. However, several studies have demonstrated recoverable *GBA1*-associated lipid alterations, including the accumulation of GCase substrates (glucosylceramide, glucosylsphingosine) and other disruptions within sphingolipid metabolism pathways [1] and the accumulation of monounsaturated lipid species [2] across various model systems. That said, the accumulation of glucosylceramide *has* been repeatedly demonstrated specifically in patient-derived iPSC midbrain dopaminergic neurons carrying either *GBA1* L444P or N370S mutations [see 4 for example]. Since lipid droplet accumulation is a more general indicator of lipid dyshomeostasis that is easily observed (via ICC), I have opted for that instead.

Method: LC/MS

Sources cited: [1] Navarro-Romero et al., 2022; [2] Peng et al., 2021; [3] Nuber et al., 2025; [4] Schondorf et al., 2014.